

SKILL – 11

PART 1 — Command History

Code:

```
GNU nano 7.2 skill11_history.c
#include <stdio.h>
#include <string.h>

#define MAX_HISTORY 5
#define MAX_CMD_LEN 100

char history[MAX_HISTORY][MAX_CMD_LEN];
int history_count = 0;

void add_command(char *cmd)
{
    if (history_count < MAX_HISTORY)
    {
        strcpy(history[history_count], cmd);
        history_count++;
    }
    else
    {
        for (int i = 1; i < MAX_HISTORY; i++)
        {
            strcpy(history[i - 1], history[i]);
        }
        strcpy(history[MAX_HISTORY - 1], cmd);
    }
}

[ Read 78 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^_A Set Mark
^X Exit      ^R Read File  ^_ Replace    ^U Paste      ^J Justify    ^_ Go To Line ^_B Redo      ^_G Copy
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-11$ gcc skill11_history.c -o skill11_history
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-11$ ./skill11_history
OS Skill-11: Command History Management

Command History:
1. pwd
2. gcc program.c
3. ./program
4. git status
5. whoami

Retrieved command: ./program

History validation: CONSISTENT

History capacity: 5 commands
```

PART 2 — Pipeline Management

Code:

```
GNU nano 7.2 skill11_pipeline.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>

#define MAX_COMMANDS 2

void validate_pipeline(char *commands[])
{
    printf("Validating pipeline...\n");
    for (int i = 0; i < MAX_COMMANDS; i++)
    {
        if (commands[i] == NULL || strlen(commands[i]) == 0)
        {
            printf("Pipeline validation: FAILED\n");
            return;
        }
    }
    printf("Pipeline validation: SUCCESS\n");
}

int main()
{
    [ Read 103 lines ]
```

Output:

```
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-11$ gcc skill11_pipeline.c -o skill11_pipeline
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-11$ ./skill11_pipeline
OS Skill-11: Pipeline Management

Pipeline commands:
1. ls
2. wc -l

Execution order:
Process 1 -> Process 2
Validating pipeline...
Pipeline validation: SUCCESS
5
Pipeline execution completed successfully.
```

Check both files:

```
Pipeline execution completed successfully.
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-11$ ls -l skill11_*
-rwxr-xr-x 1 bhardwaj bhardwaj 16280 Oct  7 14:14 skill11_history
-rw-r--r-- 1 bhardwaj bhardwaj  1463 Oct  7 14:13 skill11_history.c
-rwxr-xr-x 1 bhardwaj bhardwaj 16440 Oct  7 14:15 skill11_pipeline
-rw-r--r-- 1 bhardwaj bhardwaj  1737 Oct  7 14:15 skill11_pipeline.c
bhardwaj@Pranathi:~/2520090182_OSSP/2520090182_Skill/Skill-11$ |
```